Graph Class - 5

Special class

Shortest path

sssp → bfs

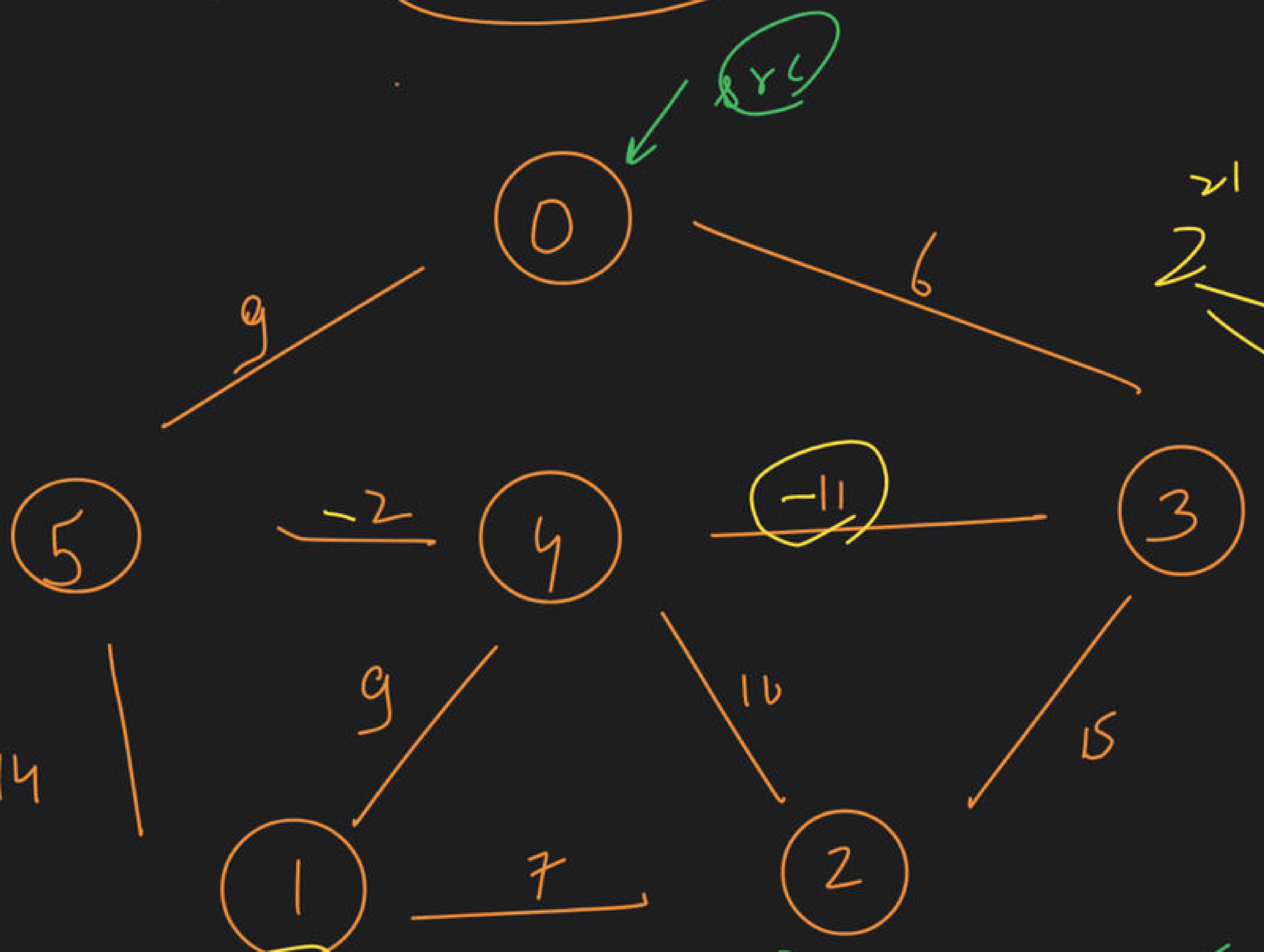
sssp → dfs

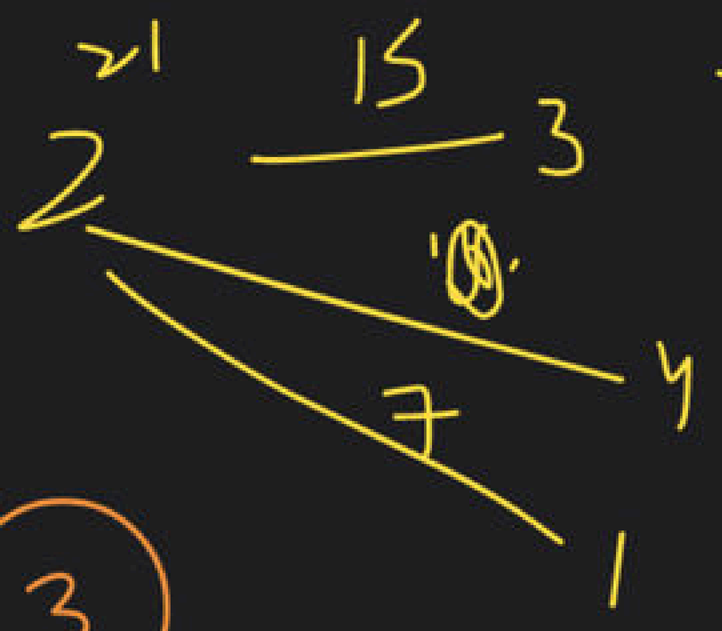
Love Babbar • Oct 3, 2024

→ Dijkstra

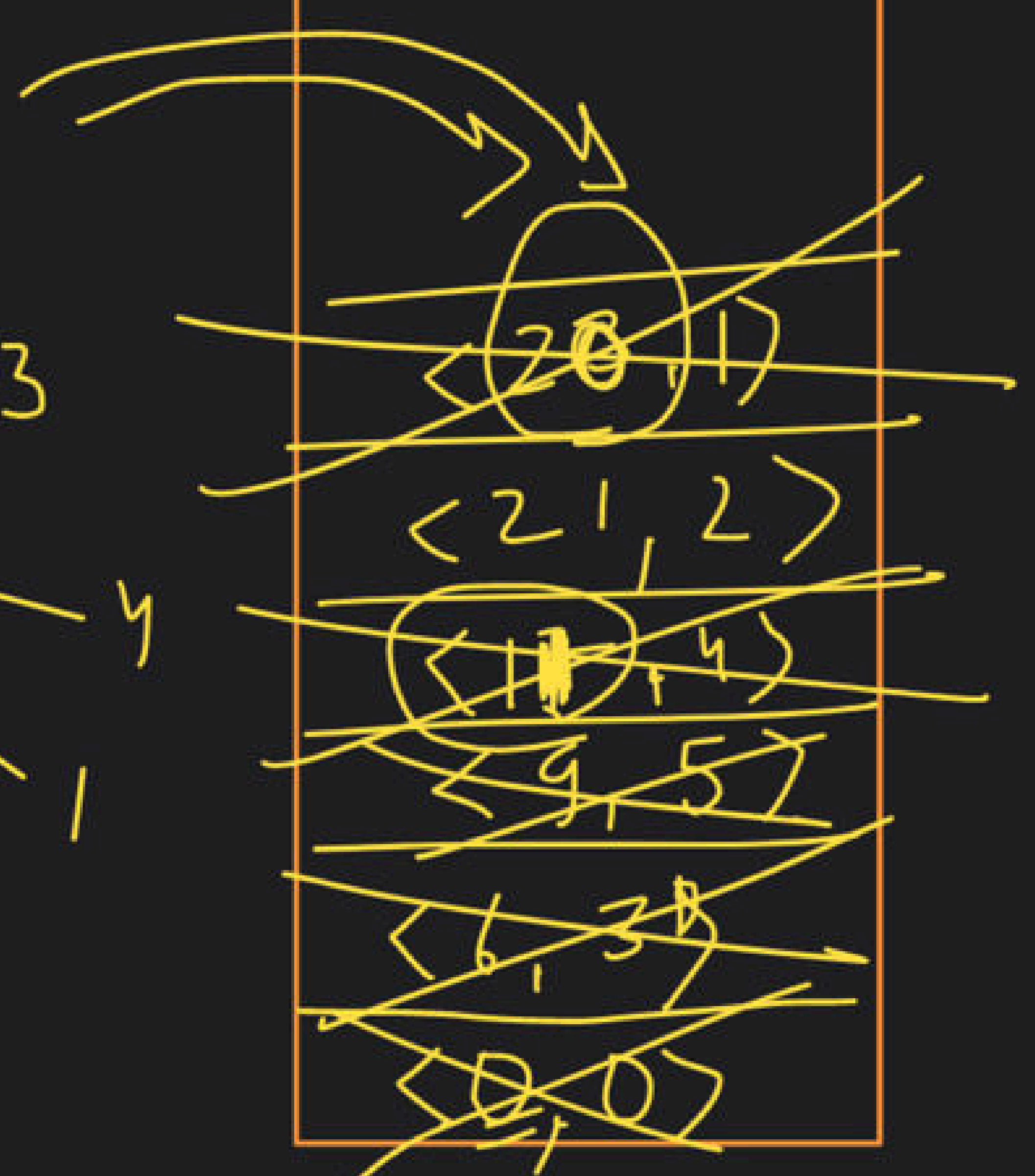
Algo: Greedy → P.Q
→ set

sssp





set



pair<int, int>
↓ dist
↓ node

initial state
↓
dist(src) = 0

	0	1	2	3	4	5
dist	0	20	21	6	11 ~~17~~	9
	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~

-ve wt

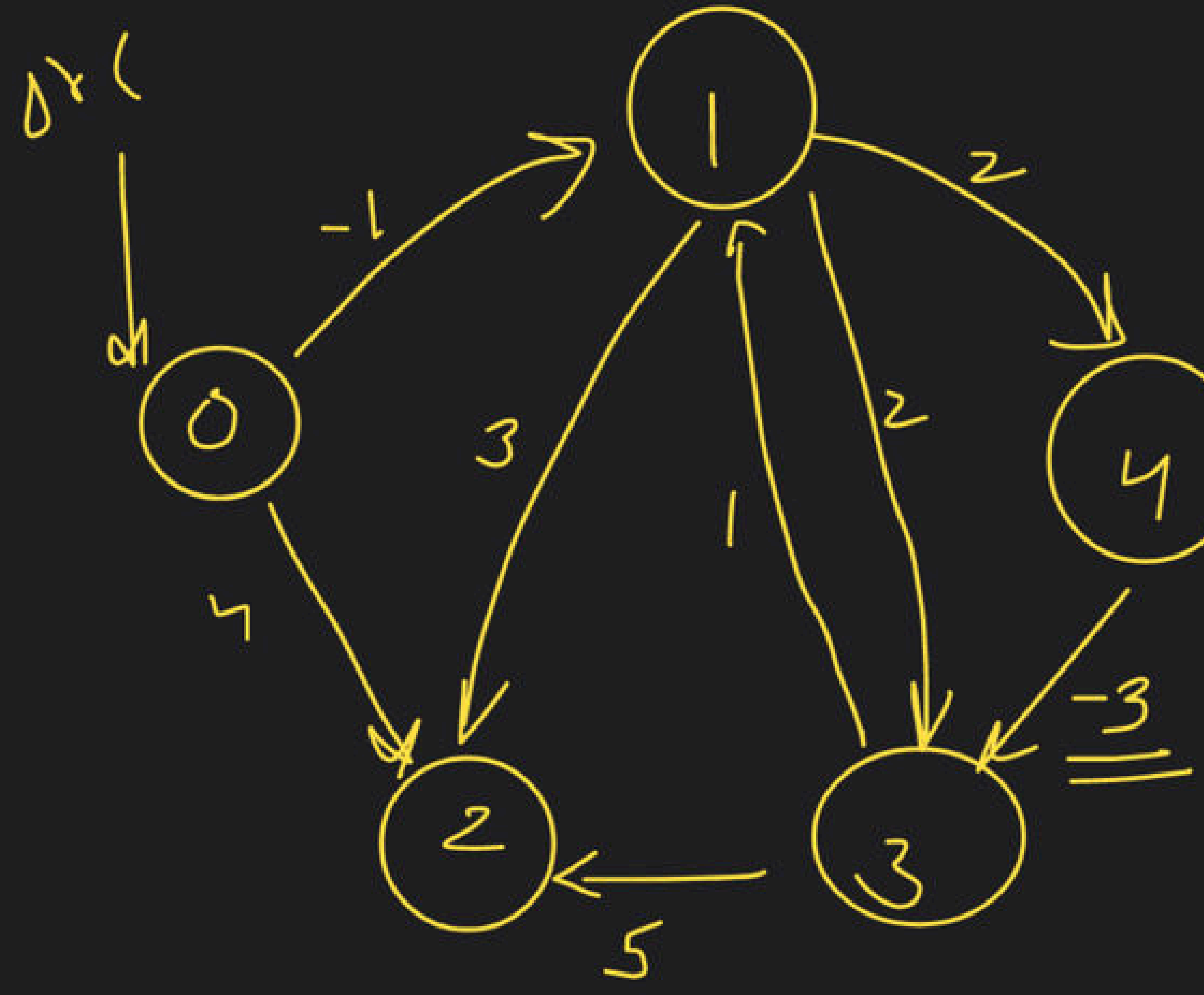
-ve cycle

$\propto$

→ Bellman Ford → △

Relaxation → (N - 1) times → why → ?

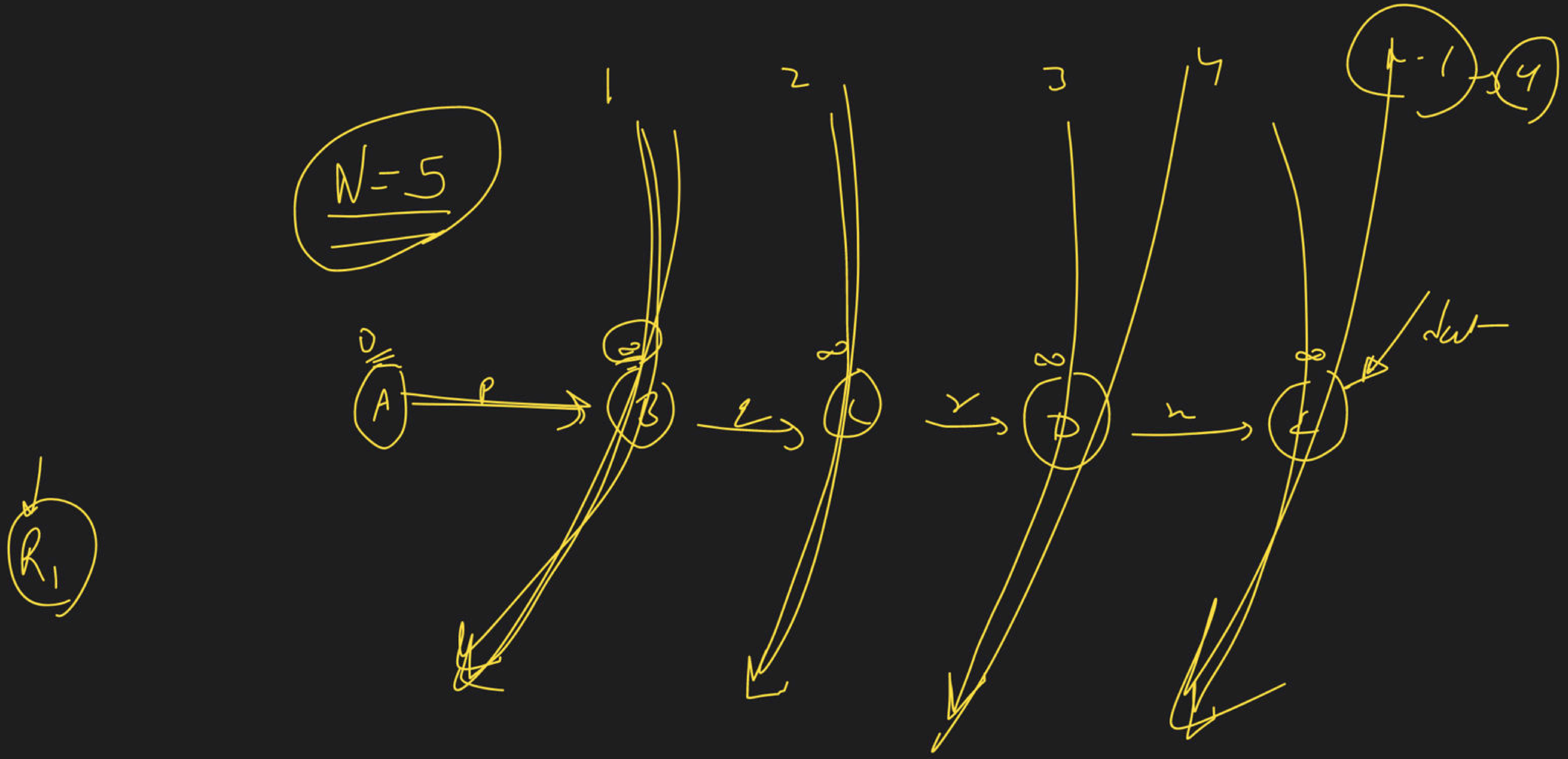
Result → shortest distance

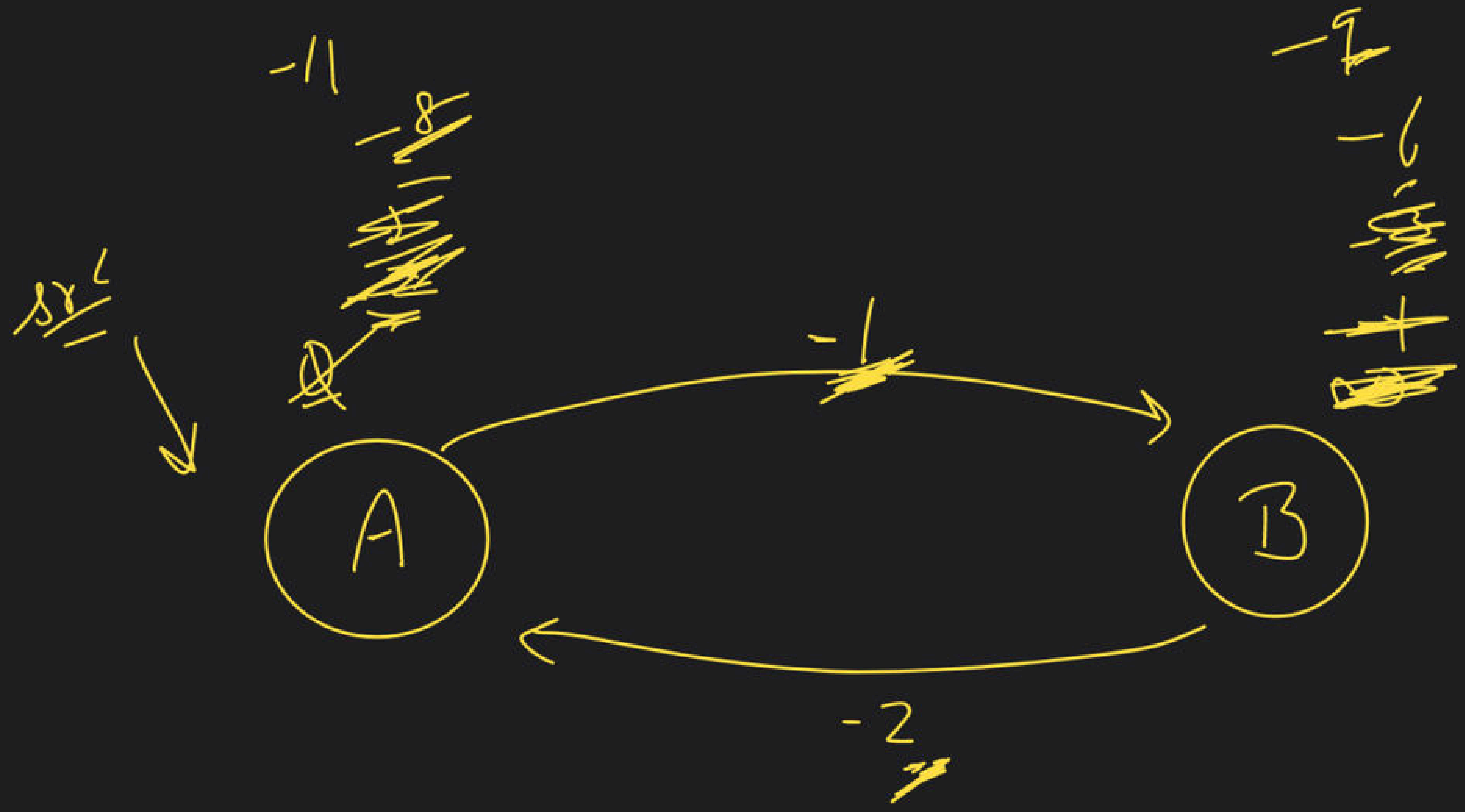


$$dist[u] + wt < dist[v]$$

→ update

N = 5





R_1

R_2

R_3

R_1

-ve cycle → Relaxation → $(N-1)$ time

↓

update ← Ihard

Edge List

$0 \xrightarrow{-1} 1$

$1 \xrightarrow{2} 4$

$0 \xrightarrow{4} 2$

$3 \xrightarrow{5} 2$

$4 \xrightarrow{-3} 3$

$1 \xrightarrow{3} 2$

$1 \xrightarrow{2} 3$

$3 \xrightarrow{1} 1$

N or V = 5

Relaxation #1

0 ~~∞~~	-1 ~~∞~~	2 ~~4~~ ~~∞~~	-2 ~~∞~~	1 ~~∞~~
0	1	2	3	4

Relaxation #2

0	-1	2	-2	1
0	1	2	3	4

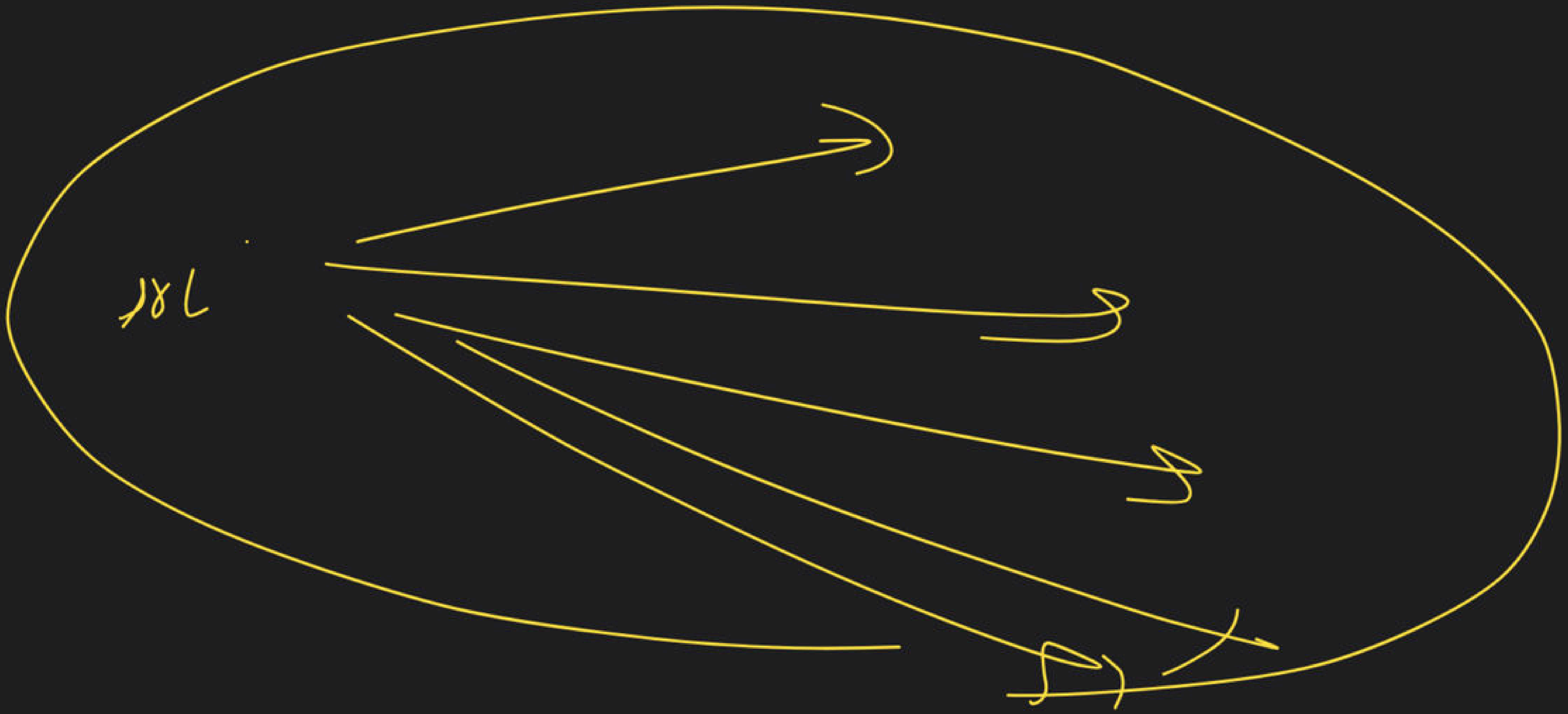
$\underbrace{\qquad\qquad\qquad\qquad}_{\text{No change}}$

break

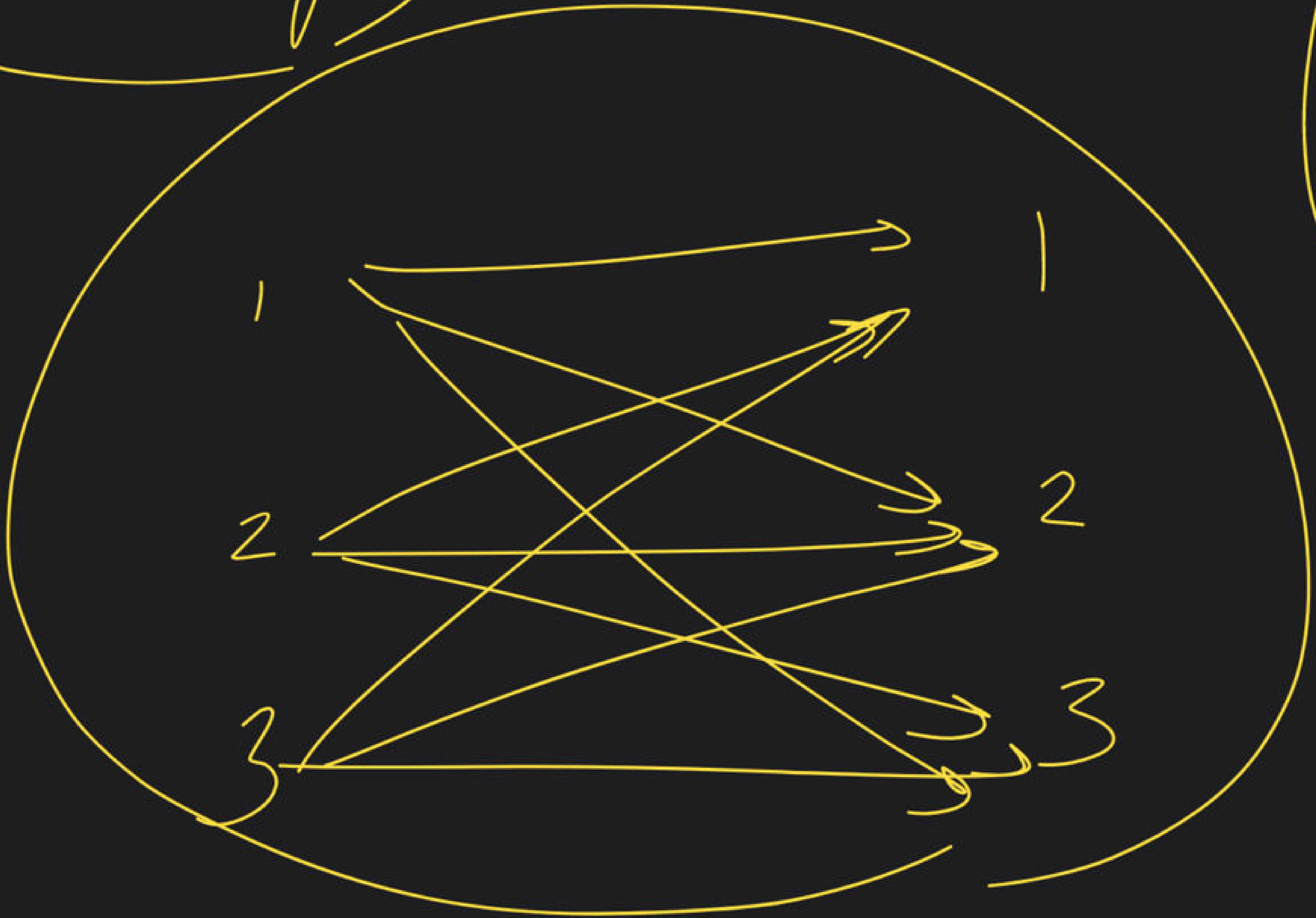
#3 → No change

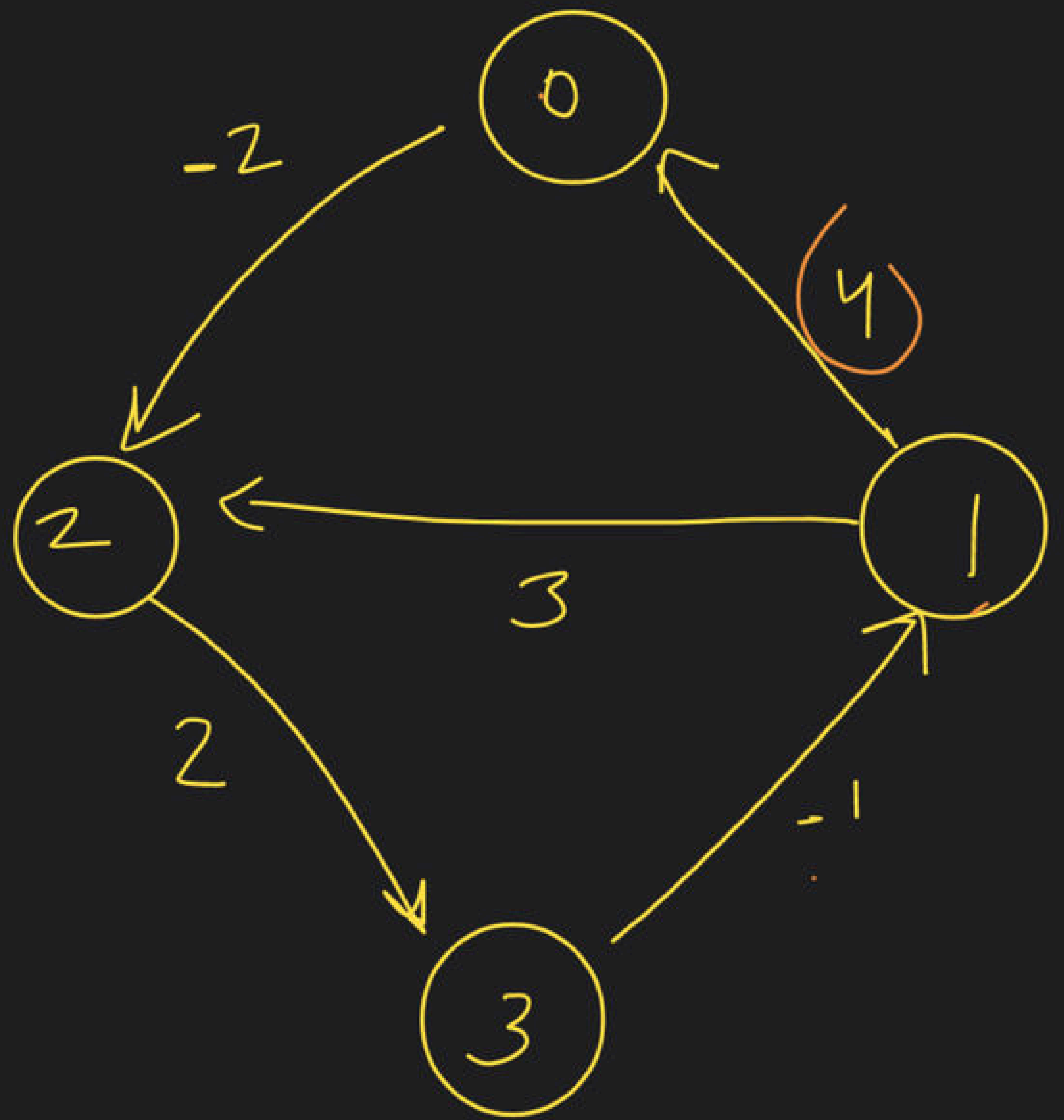
#4 → No change

SSSP



msp → floyd warshal algo





① 1, 1, 1 → 0

① grab-dist → copy

③ return

dist[u][v] = min → dist[u][v]

dist[u][k] + dist[k][v]

	0	1	2	3
0	0	-1	-2	0
1	4	0	3	4
2	5	1	0	2
3	3	-1	1	0

for(k → 0 → n)
{ for(src → 0 → n)
} for(dst → 0 → n) → logic

O(n^3)